

A physics-informed digital twin for tube-life-aware operation of industrial steam methane reformers under uncertainty

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Abstract

Background. Reformer tubes are the life-limiting component of a steam methane reformer, yet operating decisions are rarely taken with a quantified life cost. **Methods.** We couple a one-dimensional Xu–Froment tube model, a long-furnace model of a top-fired box furnace, a Larson–Miller creep-life module and Gaussian-process surrogates with conformal prediction into a digital twin. The tube model is verified against the Xu–Froment industrial case, the coupled model is calibrated on three published plant operating points with leave-one-out cross-validation, and the twin is exercised through global sensitivity analysis, hourly annual scenarios, multi-objective optimisation and Monte Carlo uncertainty propagation. **Results.** Calibration reproduces outlet temperatures within 4K and wall temperatures within 12K. Firing intensity, steam-to-carbon ratio and excess air explain 82 % of the hot-spot temperature variance. Load-following costs less than 2 % extra life per kmol of hydrogen, whereas catalyst ageing under a constant methane-slip target multiplies the life cost by 2.25 in four years. A Pareto knee delivers +16 % hydrogen at -18 % fuel and 0.28 of the base life cost, and creep-curve scatter contributes 62 % of the life-prediction variance. **Conclusions.** The twin ranks operating regimes robustly, but absolute lives require alloy-specific rupture data and a measured flame-length response to load.

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1. Introduction

Steam methane reforming (SMR) still supplies most of the world’s merchant and captive hydrogen, and it will remain the dominant route for at least a decade while electrolytic capacity is built up [TODO: support with a current IEA or similar figure and reference]. The core of an SMR plant is a top-fired or side-fired furnace in which several hundred centrifugally cast high-alloy tubes, filled with nickel catalyst, carry a mixture of natural gas and steam at 25–35 bar while their outer walls are heated to 1100–1200 K. The tubes operate in the creep regime of the alloy by design, and their remaining life is the single most important maintenance variable of the plant: a tube failure forces an unplanned shutdown, and a full retube costs of the order of [TODO: quote a retube cost range with a reference]. Tube life is in turn governed almost entirely by the outer-wall temperature at the hottest elevation, because the rupture time of the alloy roughly halves for every 10–15 K of additional metal temperature [TODO: confirm the rule of thumb for HP-Nb alloys and cite].

Operating decisions therefore trade hydrogen production, fuel consumption and tube life against each other, but this trade-off is rarely quantified. Plants are usually run against a maximum tube-wall temperature limit read from pyrometer surveys, and the effect of a load change, a steam-to-carbon change or a burner adjustment on the tube life is left to experience. Two recent developments make a quantitative approach both necessary and possible. On the demand side, hydrogen plants are increasingly expected to follow variable downstream demand or to operate alongside intermittent renewable hydrogen, which raises the question of how much tube life flexible operation actually costs. On the modelling side, validated first-principles models of the reformer tube [1, 2, 3, 4] and of the furnace [5, 6] exist, machine-learning surrogates make such models fast enough for hourly simulation and optimisation [7, 8, 9, 10], and creep data for the relevant alloys are available in public databases [11, 12]. What is missing is a coupled, calibrated and uncertainty-aware chain from operating inputs to tube life, built entirely on published data and open code, so that its conclusions can be checked and its data replaced.

34 *1.1. Previous work*

35 The reformer tube has been modelled at every level of complexity. Xu
36 and Froment [1] established the Langmuir–Hinshelwood–Hougen–Watson ki-
37 netics of methane steam reforming and water-gas shift on a Ni/MgAl₂O₄
38 catalyst that remains the standard intrinsic model, and in the companion
39 paper [2] they simulated an industrial tube with a one-dimensional hetero-
40 geneous model and reported axial profiles of conversion, temperature and
41 pressure for one operating point. Plehiers and Froment [3] coupled the tube
42 to a zone model of a top-fired furnace, and Pantoleontos et al. [4] revisited
43 the heterogeneous tube model with a dynamic formulation. Latham and co-
44 workers [13, 5] built a three-dimensional furnace model of a top-fired box
45 reformer, calibrated it on plant measurements including infrared tube-wall
46 temperature surveys, and made the plant operating points and tube-wall
47 temperatures available in the thesis appendix [13]; that data set is the basis
48 of the present calibration. Zamaniyan et al. [6] proposed a simpler furnace
49 model for control-oriented use.

50 On the materials side, the creep behaviour of the HP-Nb family of cen-
51 trifugally cast tube alloys has been characterised in the long-term data sheets
52 of the National Institute for Materials Science [11, 12] and in remaining-life
53 studies of service-exposed tubes [14, 15, 16, 17]. Yeh [18] combined a tube-
54 wall temperature estimate with a manufacturer’s Larson–Miller curve for a
55 proprietary HP-Nb alloy to estimate life, and the same curve is used here as
56 a placeholder until the public data sheets are transcribed. The Larson–Miller
57 parameter [19] and the linear life-fraction rule of Robinson [20] remain the
58 basis of design codes such as API 530 [21].

59 Data-driven and hybrid approaches have been applied to reformers in
60 the last decade: neural-network models of the reformer [7, 8], optimisation of
61 reformer operation and furnace balancing [9, 10], and, more recently, physics-
62 informed and digital-twin formulations [22, 23, 24]. These studies generally
63 address production and efficiency; tube life, when it appears, is a constraint
64 on a maximum temperature rather than an objective with a quantified cost,
65 and uncertainty in the kinetics, the heat transfer and the creep data is not
66 propagated to the conclusions.

67 *1.2. Contribution and research questions*

68 This paper presents a physics-informed digital twin of an industrial top-
69 fired SMR that links operating inputs to tube creep life through a one-
70 dimensional tube model, a long-furnace model, a Larson–Miller life module,

71 machine-learning surrogates with distribution-free prediction intervals, and
 72 Monte Carlo uncertainty propagation (Fig. 1). The tube model is verified
 73 against the industrial case of Xu and Froment [2], the coupled model is cal-
 74 ibrated on the plant data of Latham [13] with cross-validation, and every
 75 number in the results is produced by a versioned open pipeline from the
 76 published data. The twin is then used to answer four research questions:

77 RQ1 Which operating variables govern the hot-spot temperature and the
 78 creep-life consumption of the tube, and how do they interact?

79 RQ2 What is the tube-life cost, per unit of hydrogen produced, of flexible
 80 operation (load-following, renewable-following, over-firing events) and
 81 of catalyst ageing, relative to steady operation?

82 RQ3 Which operating regimes are Pareto-optimal in hydrogen production,
 83 fuel consumption and life consumption, and how does the answer de-
 84 pend on how steam is charged?

85 RQ4 How uncertain are these answers, which uncertainty sources dominate,
 86 and which conclusions survive the uncertainty?

87 Section 2 describes the models, the data and the numerical methods. Sec-
 88 tion 3 reports verification, calibration and the answers to RQ1–RQ4. Sec-
 89 tion 4 draws the industrial implications and states the limitations, and Sec-
 90 tion 5 concludes.

91 2. Methods

92 The twin is implemented as a Python package (`rdt`) whose modules follow
 93 the chain of Fig. 1b. All physical data (kinetic parameters, plant operating
 94 points, creep curves) are read from annotated YAML and CSV files; no
 95 number is hard-coded in the model code. Thermodynamic properties, heat
 96 capacities and reaction equilibrium constants are taken from Cantera [25]
 97 with the GRI-Mech 3.0 species data so that kinetics and energy balance are
 98 mutually consistent.

99 2.1. Reactor tube model

100 The tube is described by a steady one-dimensional pseudo-homogeneous
 101 plug-flow model with axial coordinate z from the inlet (top) to the outlet.
 102 The state consists of the molar flows F_i of CH_4 , H_2O , H_2 , CO , CO_2 and N_2 ,
 103 the process-gas temperature T and the pressure p . The intrinsic kinetics are

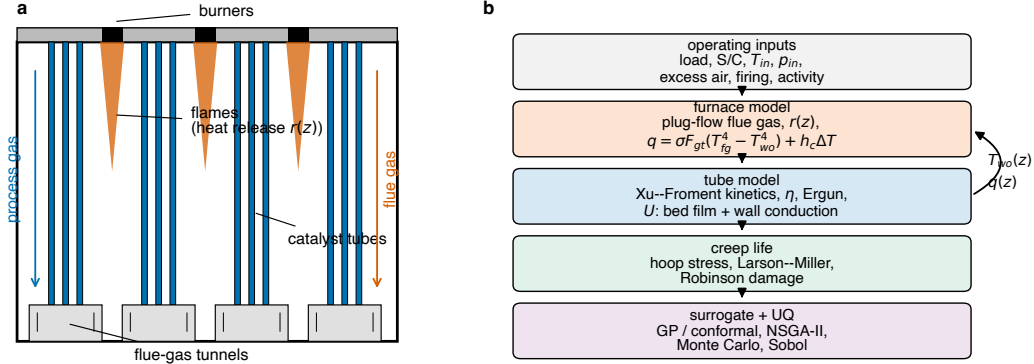


Figure 1: (a) Cross-section of a top-fired box reformer: burners in the roof between rows of catalyst tubes, flames releasing heat in the upper part of the firebox, flue gas flowing downward co-currently with the process gas and leaving through tunnels at the floor. (b) Module chain of the digital twin: operating inputs are mapped through the furnace and tube models, which exchange the outer-wall temperature profile and the heat flux, to the creep-life module; surrogates and uncertainty quantification wrap the chain.

those of Xu and Froment [1] for steam reforming (r1), water-gas shift (r2)
and overall reforming to CO_2 (r3):

$$r_1 = \frac{k_1}{p_{\text{H}_2}^{2.5}} \frac{p_{\text{CH}_4} p_{\text{H}_2\text{O}} - p_{\text{H}_2}^3 p_{\text{CO}} / K_1}{\text{DEN}^2}, \quad (1)$$

$$r_2 = \frac{k_2}{p_{\text{H}_2}} \frac{p_{\text{CO}} p_{\text{H}_2\text{O}} - p_{\text{H}_2} p_{\text{CO}_2} / K_2}{\text{DEN}^2}, \quad (2)$$

$$r_3 = \frac{k_3}{p_{\text{H}_2}^{3.5}} \frac{p_{\text{CH}_4} p_{\text{H}_2\text{O}}^2 - p_{\text{H}_2}^4 p_{\text{CO}_2} / K_3}{\text{DEN}^2}, \quad (3)$$

$$\text{DEN} = 1 + K_{\text{CO}} p_{\text{CO}} + K_{\text{H}_2} p_{\text{H}_2} + K_{\text{CH}_4} p_{\text{CH}_4} + K_{\text{H}_2\text{O}} p_{\text{H}_2\text{O}} / p_{\text{H}_2}, \quad (4)$$

with partial pressures in bar. The rate coefficients k_j and adsorption constants K_i follow the Arrhenius and van't Hoff forms with the pre-exponential factors, activation energies and adsorption enthalpies of Table 5 of [1], re-parameterised at the reference temperatures 648 K and 823 K used in that table so that the reported confidence intervals can be propagated directly (Section 2.7). The equilibrium constants K_1 – K_3 are computed from Gibbs energies of formation rather than from the fitted expressions of the original paper, which removes a source of inconsistency with the energy balance. A floor of 10^{-6} bar on p_{H_2} avoids the singularity of the rate expressions at a hydrogen-free inlet.

116 The balances per tube are

$$\frac{dF_i}{dz} = A_{cs} \rho_b \sum_j \nu_{ij} \eta_j r_j, \quad (5)$$

$$\sum_i F_i c_{p,i} \frac{dT}{dz} = \pi d_i U (T_{wo}(z) - T) - A_{cs} \rho_b \sum_j \eta_j r_j \Delta H_j(T), \quad (6)$$

$$-\frac{dp}{dz} = \frac{1-\phi}{\phi^3} \left(1.75 + 150 \frac{1-\phi}{Re_p} \right) \frac{\rho_g v_s^2}{d_p}, \quad (7)$$

117 where A_{cs} is the cross-section, ρ_b the bed density, ν_{ij} the stoichiometric co-
 118 efficients, η_j the effectiveness factors, ϕ the bed voidage, d_p the equivalent
 119 particle diameter, v_s the superficial velocity and Re_p the particle Reynolds
 120 number; Eq. (7) is the Ergun equation [26]. Intraparticle diffusion limitation
 121 is severe in reformer catalysts, and rather than solving the pellet problem we
 122 use the constant effectiveness factors adopted by Latham et al. [5]: $\eta_j = 0.1$
 123 along the tube and 0.05 in the first 10 % of the length, where the feed contains
 124 no hydrogen and the rate expressions are stiff [TODO: confirm the 0.05/0.1
 125 profile and its justification against Latham et al. (2011)]. Higher alkanes in
 126 the feed (C₂–C₅) are converted to CO and H₂ at the inlet by the rule used
 127 by Latham [13] [TODO: state the exact inlet conversion rule and confirm its
 128 stoichiometry].

129 The overall heat-transfer coefficient referred to the inner surface combines
 130 the bed-side film coefficient α_i and the conduction through the tube wall,

$$\frac{1}{U} = \frac{1}{\alpha_i} + \frac{d_i}{2\lambda_t} \ln \frac{d_o}{d_i}, \quad T_{wi} = T_{wo} - q_o'' \frac{d_o}{2\lambda_t} \ln \frac{d_o}{d_i}, \quad (8)$$

131 with tube conductivity $\lambda_t = 29.6 \text{ W m}^{-1} \text{ K}^{-1}$ from [13]. The film coefficient
 132 uses the Leva–Grummer correlation [27] in the form written by Latham et
 133 al. [5], multiplied by a heat-transfer factor f_{htg} which is estimated from plant
 134 data (Section 2.5). The alternative chain of Xu and Froment [2], based on
 135 De Wasch and Froment [28] and Kunii and Smith [29], is implemented as
 136 an option and discussed in Section 3.1. Equations (5)–(7) are integrated
 137 with the LSODA solver of SciPy for a prescribed outer-wall profile $T_{wo}(z)$
 138 (stand-alone mode) or as part of the coupled system below.

139 2.2. Furnace model and coupling

140 The furnace is a top-fired box with rows of burners in the roof between
 141 rows of tubes (Fig. 1a; Latham’s geometry has 7 rows of 48 tubes and 8 rows

of 12 burners [13]). Instead of the zone method of Hottel and Sarofim [30] used by Plehiers and Froment [3], we use a one-dimensional “long-furnace” model in the sense of Roesler [31]: the flue gas is a plug flow travelling downward co-currently with the process gas, and the radiative exchange between the gas and the tube surface is lumped into one effective exchange factor F_{gt} . Per unit height z (zero at the roof),

$$\dot{n}_{fg} c_{p,fg} \frac{dT_{fg}}{dz} = r(z) - q(z) \pi d_o N_t, \quad (9)$$

$$q(z) = \sigma F_{gt} (T_{fg}^4 - T_{wo}^4) + h_c (T_{fg} - T_{wo}), \quad (10)$$

$$q(z) = U (T_{wo} - T) \frac{d_i}{d_o}, \quad (11)$$

where N_t is the number of tubes, q the flux per unit outer tube area and h_c a convective coefficient from a flat-plate correlation [TODO: state the convective correlation and its magnitude relative to radiation]. The heat-release density $r(z)$ is a downward-opening parabola that is zero at $z = L_q L$, releases the fraction α_{top} of the heat in the top fifteenth of the furnace height and integrates to $(1 - f_{loss}) Q_{comb}$, with $f_{loss} = 0.02$ for casing losses; this is the continuous form of the sectional heat-release profile of Latham [13]. The dimensionless flame length L_q and α_{top} are calibration parameters. Because a pure plug flow starts at the adiabatic flame temperature of the fuel mixture and would give an unphysical cold inlet region before the first heat release, the top zone is treated as well mixed: the fraction α_{top} of the heat is released instantly at the roof. The combustion heat Q_{comb} and the flue-gas flow and composition follow from the lower heating value and complete combustion of the fuel gas and pressure-swing-adsorption off-gas with the specified excess air.

Equation (11) closes the system: at every z the outer-wall temperature T_{wo} is the root of the flux balance between Eqs. (10) and (11), found by a bracketed root search inside the right-hand side of the ODE system, so that the furnace and tube equations are integrated together in one downward sweep. The global energy closure of the coupled solution is verified to 10^{-5} of the combustion heat.

2.3. Tube wall and creep life

The creep-life module converts the outer-wall temperature profile into a life estimate. The hoop stress is computed from the mean-diameter formula

172 of API 530 [21], $\sigma_h = p(d_o - t)/(2t)$ with wall thickness t , evaluated at the
 173 pressure of the process gas at each elevation; thin-wall and Lamé variants
 174 are implemented for comparison and differ by less than [TODO: quote the
 175 difference] at the tube geometry used. The wall thickness is not given by
 176 Latham and is taken as 15 mm [TODO: confirm a typical wall thickness for a
 177 12.5 m HP-Nb tube of this diameter] and varied in the uncertainty analysis.
 178 The temperature entering the life calculation is the outer-wall temperature,
 179 which is conservative with respect to the mid-wall temperature; the difference
 180 is shown in Fig. 4.

181 The rupture time follows the Larson–Miller parameter [19],

$$\text{LMP} = 10^{-3} T (C + \log_{10} t_r), \quad \log_{10} \sigma = P_3(\text{LMP}), \quad (12)$$

182 with T in K, t_r in h, $C = 22.96$ and a cubic master curve P_3 fitted to the
 183 minimum-property rupture points of the manufacturer’s curve for Manau-
 184 rite XM reproduced by Yeh [18]. This curve is a *placeholder*: the alloy of
 185 the Latham plant is not stated, and the manufacturer’s curve carries no
 186 scatter information. The module is written so that the curve can be re-
 187 placed by a heat-normalised fit to the NIMS creep data sheets for the 25Cr–
 188 35Ni–Nb alloys [11, 12] (Section 2.4). Damage under a piecewise-constant
 189 history is accumulated with the linear life-fraction rule of Robinson [20],
 190 $D = \sum_k \Delta t_k / t_r(T_k, \sigma_k)$, and rupture is predicted at $D = 1$. Throughout
 191 the paper, life is reported as a *consumption rate* $1/t_r$ at the hot spot, or as
 192 damage per kmol of hydrogen, relative to the calibrated base case, because
 193 ratios of rupture times are far less sensitive to the placeholder curve than
 194 absolute values are.

195 2.4. Data

196 *Verification data.* The industrial simulation of Xu and Froment [2] (Fig. 3 of
 197 that paper) provides axial profiles of methane and CO₂ conversion, process-
 198 gas, inner-wall and outer-wall temperatures and pressure for one operating
 199 point with the feed of their Table 2. The curves were digitised from the
 200 published figure [TODO: name the digitising tool and the resolution] and
 201 stored as tidy CSV files with the digitisation metadata (Supplementary Sec-
 202 tion S1). The outer-wall curve serves as the boundary condition and the
 203 remaining curves as targets.

204 *Plant data.* Latham [13] reports, in Appendix H, four operating points of an
 205 industrial top-fired reformer (Plants A, B and two states of Plant C) with

inlet conditions, feed and fuel compositions, outlet temperature and composition, flue-gas outlet temperature and the tube-wall temperatures measured by infrared pyrometer at two peep-hole elevations (3.66 and 8.53 m from the top of the 12.5 m heated length). Table 2 lists the cases. The stated measurement uncertainties are 2 K for the outlet temperature, 3 K for the wall temperatures and 8 K for the flue gas [TODO: confirm these uncertainties against the thesis and whether they are one-sigma]. Plant C1 is excluded from calibration because its feed flow is inconsistent (equal to the air flow and 22 % larger than Plant A at a lower rate) and is likely a transcription error in the source [TODO: confirm the exclusion of Plant C1]. The excess air of each case is derived from the reported fuel, off-gas and air flows and is 21.6 % for Plant A.

Creep data. The placeholder Larson–Miller curve is transcribed from Yeh [18]. A transcription template and a schema for the NIMS data sheets are provided, and an ingestion routine fits C and the master curve by cross-validation with per-heat offsets; it is validated on a synthetic data set with known C and scatter (recovered within 10 %) but has not yet been applied to real NIMS data, which must be transcribed from the printed sheets [TODO: state the status of the NIMS transcription at submission].

2.5. Calibration and cross-validation

Four parameters are uncertain in the coupled model and cannot be taken from the literature: the exchange factor F_{gt} , the flame length L_q , the top-zone release fraction α_{top} and the heat-transfer factor f_{htg} . They are estimated by weighted non-linear least squares on the three plant cases, with residuals in the outlet temperature, the outlet pressure, the outlet wet composition, the flue-gas outlet temperature and the two wall temperatures (nine residuals per case), each scaled by its measurement uncertainty. The trust-region-reflective algorithm with bounds is started from a Latin-hypercube set of initial points to guard against local minima, and the parameter covariance is obtained from the Jacobian at the optimum with a t -based 95 % interval. Predictive performance is assessed by leave-one-case-out cross-validation: the parameters are refitted on two cases and the held-out case is predicted. The four-parameter fit places α_{top} at its bound with a correlation of 0.985 to L_q , so the adopted configuration fixes $\alpha_{top} = 0.182$ [TODO: state the origin of the value 0.182] and estimates the remaining three; the increase in χ^2 is reported in Section 3.1.

242 2.6. Surrogates and conformal prediction

243 The coupled model takes 1–3 s per operating point, which is adequate
 244 for calibration but not for hourly annual histories, evolutionary optimisa-
 245 tion or Monte Carlo. Surrogates are trained on a Latin-hypercube design of
 246 2000 points over the seven operating inputs (Table S1 of the Supplementary
 247 material) [TODO: decide whether the ranges table goes in the main text
 248 or the supplement] for eight outputs (hot-spot temperature $T_{wo,max}$ and its
 249 position, outlet temperature and pressure, dry methane slip, net hydrogen,
 250 duty, and $\log_{10} t_r$ at the hot spot). Three model families are compared with
 251 scikit-learn [32]: Gaussian-process regression with a Matérn 5/2 kernel on
 252 a 1000-point subset, histogram gradient boosting and a multilayer percep-
 253 tron, each with and without two physics-derived features (the equilibrium
 254 methane conversion at outlet conditions and the specific duty). The model
 255 with the lowest five-fold cross-validated error is selected per target and eval-
 256 uated on an independent Saltelli design of 4096 points. Prediction intervals
 257 are obtained by the cross-conformal (CV+) method [33] as implemented in
 258 MAPIE [34], which provides finite-sample marginal coverage without a dis-
 259 tributional assumption; the Gaussian-process native intervals are reported
 260 for comparison.

261 2.7. Sensitivity, scenarios, optimisation and uncertainty propagation

262 *RQ1.* Global sensitivity uses the variance-based Sobol indices [35] estimated
 263 with SALib [36] on the Saltelli design evaluated with the physics model,
 264 together with an open-loop Latin-hypercube campaign and a closed-loop grid
 265 in steam-to-carbon ratio and load in which the firing is solved so that the
 266 outlet temperature equals the Plant A value (1105.55 K).

267 *RQ2.* Annual histories of 8760 hourly quasi-steady states are evaluated through
 268 the surrogates, with every 24th hour and every hour outside the training
 269 bounds re-evaluated by the physics model. Two control modes are repre-
 270 sented by inverting the surrogates for the firing factor with a vectorised bi-
 271 section: holding the outlet temperature at its base value, or holding the dry
 272 methane slip. The scenarios are S1 steady base; S2 daily load-following (full
 273 load 06–22 h, 70 % at night, ramps of 0.10 per hour); S3 renewable-following
 274 (a bounded Ornstein–Uhlenbeck load process between 0.60 and 1.10 with
 275 mean 0.85); S4 over-firing events (24 events per year of 10 % firing for 2 h,
 276 six events of 25 % for 1 h, and a flame-impingement proxy adding 40 K to
 277 the hot spot for 24 h per year); S5 catalyst ageing, with the activity falling

linearly from 0.30 to 0.10 over four years under each control mode; and S6 steady operation at steam-to-carbon 2.5 and 3.5. Damage is accumulated hourly by Eq. (12) and Robinson’s rule, and each scenario is also evaluated at its annual-mean inputs to quantify the error of a steady-state assessment. Start-up and shutdown transients, thermal fatigue and creep–fatigue interaction are outside the scope.

RQ3. The non-dominated sorting genetic algorithm NSGA-II [37] in pymoo [38] maximises net hydrogen per tube and minimises firebox fuel per kmol H_2 and life consumption per kmol H_2 over five decision variables (load, steam-to-carbon ratio, inlet temperature, excess air, firing) subject to methane slip not above the base value, hot-spot temperature not above 1193 K [TODO: justify the 1193 K limit, e.g. the design metal temperature of the alloy] and outlet temperature not above 1125 K, using the Gaussian-process surrogates (population 200, 300 generations). Every Pareto member is re-evaluated with the physics model and the constraints are re-checked. Because firebox fuel per kmol H_2 ignores the cost of raising the additional steam, a *steam credit* α is introduced: the total fuel counts firebox fuel plus a fraction $(1 - \alpha)$ of the heat needed to raise the process steam and preheat the feed, so that $\alpha = 1$ recovers the firebox-only objective and $\alpha = 0$ charges steam in full [TODO: confirm the steam-credit definition and the enthalpy basis for steam raising]. The iso-production front (hydrogen within 1 % of the base) is recomputed for α from 0 to 1.

RQ4. Uncertain inputs are grouped into kinetics (the six rate parameters and four adsorption constants of Xu and Froment with normal distributions from their reported 95 % intervals, a log-uniform factor of 0.5–2 on the effectiveness factors and ± 0.03 on the catalyst activity), heat transfer (the calibrated F_{gt} , L_q and f_{htg} with their joint covariance), creep (a normal scatter of 0.3 decades on $\log_{10} t_r$ representing a factor of two between heats, C uniform within ± 1 of 22.96 with the master curve refitted for each C , wall thickness 12–18 mm and tube conductivity ± 15 %) and measurement (a uniform ± 10 K offset on the pyrometer-referenced hot-spot temperature). A scrambled Sobol design of 2048 samples is propagated through the physics model at eight operating points taken from RQ2 and RQ3; the same sample is used at every point so that ratios between points are paired. Variance is decomposed by varying one group at a time (1024 samples per group) and, independently, by Sobol indices computed on a Gaussian-process surrogate of the joint sample.

314 Finally, the fraction of paired samples for which each qualitative conclusion
 315 of RQ2 and RQ3 holds is reported, both with the calibrated flame length
 316 and with a structural variant in which the flame length shrinks with load as
 317 $L_q \text{ load}^{0.5}$ [TODO: justify or replace the load exponent 0.5 of the flame-length
 318 variant]. The complete pipeline from design to figures runs in 34 minutes on
 319 a laptop.

320 3. Results

321 3.1. Verification and validation

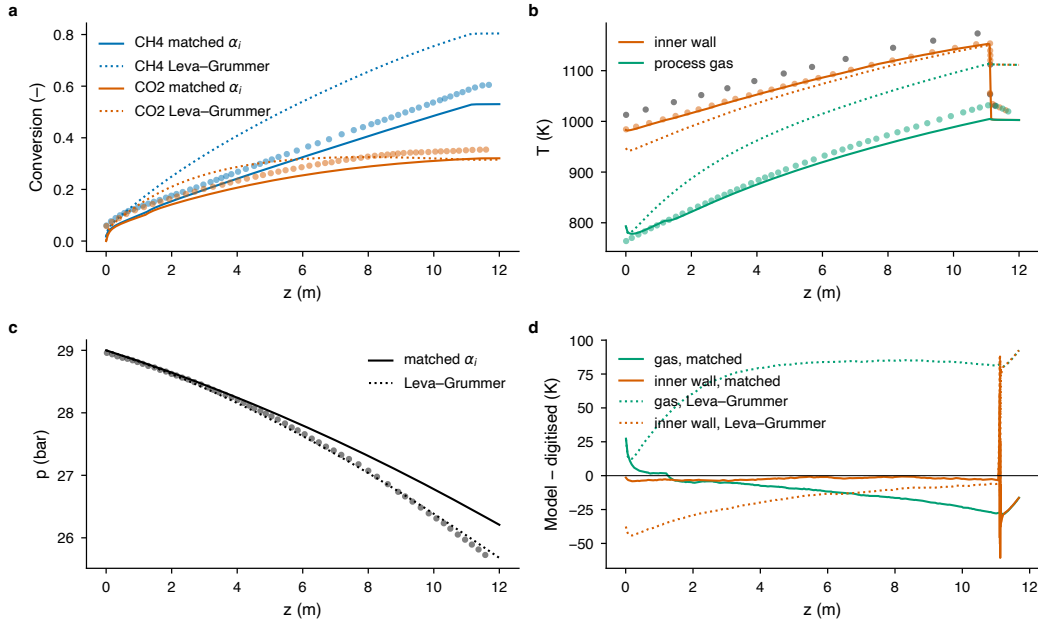


Figure 2: Verification against the digitised industrial case of Xu and Froment [2] (symbols) with the digitised outer-wall temperature as boundary condition: (a) methane and CO₂ conversion, (b) process-gas and inner-wall temperatures, (c) pressure and (d) temperature residuals, for the matched film coefficient $\alpha_i = 348 \text{ W m}^{-2} \text{ K}^{-1}$ (solid lines) and for the Leva-Grummer correlation with $f_{htg} = 1$ (dotted). Source: data/literature_validation/digitized/, xf1989_fit.yaml, v2.

322 With the digitised outer-wall temperature prescribed, the tube model's
 323 response is controlled almost entirely by the bed-side film coefficient, and
 324 the first finding of the study is that the heat-transfer chain printed in Part
 325 II of Xu and Froment [2] does not reproduce their own figure. Evaluated as

Table 1: Verification of the tube model against the digitised Fig. 3 of Xu and Froment [2]: RMSE of methane conversion x_{CH_4} , of the conversion increment $x(z) - x(0.5 \text{ m})$, of the process-gas and inner-wall temperatures ($z \geq 0.3 \text{ m}$) and of pressure. Source: `data/literature_validation/xf1989_fit.yaml` (`verification_matched_alpha`).

Configuration	RMSE x_{CH_4}	RMSE Δx	RMSE T_g (K)	RMSE T_{wi} (K)	RMSE p (bar)
Leva–Grummer, $f_{htg} = 1$, $\eta = 0.1$	0.160	nan	75.4	37.5	0.04
matched $\alpha_i = 348 \text{ W m}^{-2} \text{ K}^{-1}$, $\eta = 0.1$	0.037	0.020	13.7	16.7	0.25
matched α_i , $\eta = 0.05 / 0.1$ (adopted)	0.039	0.015	14.0	16.7	0.25
matched $\alpha_i(z)$ profile, $\eta = 0.05 / 0.1$	0.046	0.017	17.6	16.9	0.26

Back-calculated α_i : median 348, interquartile range 313–363 $\text{W m}^{-2} \text{ K}^{-1}$ over 0.5–11 m.

326 printed, the De Wasch–Froment wall coefficient with the Yagi–Kunii dynamic
327 terms gives $\alpha_i \approx 3100\text{--}3700 \text{ W m}^{-2} \text{ K}^{-1}$ at the particle Reynolds number of
328 the case, and the Leva–Grummer correlation gives $940\text{--}1580 \text{ W m}^{-2} \text{ K}^{-1}$; the
329 value back-calculated from the digitised wall and gas temperatures and the
330 wall conduction is $348 \text{ W m}^{-2} \text{ K}^{-1}$ (interquartile range 313–363). Used as
331 printed, the correlations overpredict the gas temperature by 70–115 K and
332 the conversion by 0.14–0.22 (Table 1, first row). Both dynamic terms are
333 linear in $RePr$ and were fitted at Reynolds numbers one order of magni-
334 tude lower than those of an industrial tube, so the extrapolation drives the
335 effective radial conductivity to values that no measurement supports. A dis-
336 placed decimal in the printed coefficient of the effective conductivity would
337 resolve the discrepancy, but we treat this as a hypothesis [TODO: check the
338 coefficients against De Wasch and Froment (1972) and Froment and Bischoff
339 before asserting a misprint].

340 The Xu–Froment case is therefore used as a *verification* of the tube model
341 with a matched film coefficient rather than as a validation of the heat-transfer
342 correlation. With $\alpha_i = 348 \text{ W m}^{-2} \text{ K}^{-1}$ and Latham’s inlet effectiveness
343 profile, the model reproduces the digitised profiles with an RMSE of 14.0
344 K in gas temperature, 16.7 K in inner-wall temperature and 0.015 in the
345 conversion increment along the tube (Fig. 2, Table 1); the residual offset in
346 absolute conversion of about 0.04 originates in the first half metre, where
347 the digitised curve and the inlet rule differ [TODO: comment on the inlet-
348 region offset]. This verifies kinetics, thermodynamics, the energy balance
349 and the wall conduction chain. The pressure RMSE of 0.25 bar with the
350 matched coefficient, against 0.04 bar with the Leva–Grummer run, reflects
351 the different gas temperature and hence density along the tube rather than a
352 change in the friction model [TODO: confirm this explanation of the pressure

353 residual].

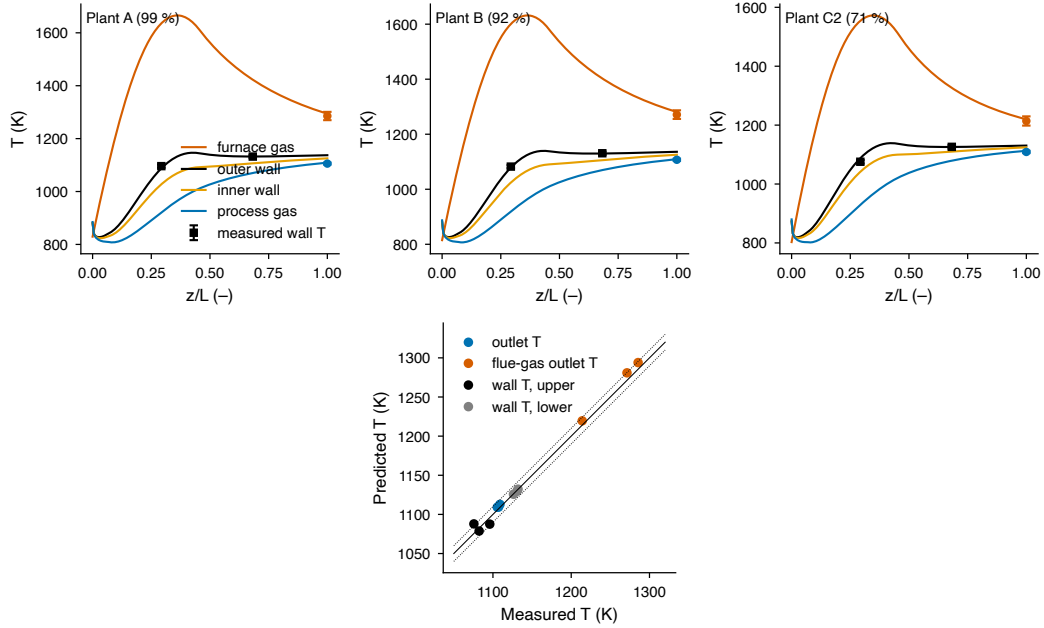


Figure 3: Coupled furnace–tube model after calibration (α_{top} fixed) on the Latham plant cases: axial profiles of furnace-gas, outer-wall, inner-wall and process-gas temperatures for Plants A, B and C2 with the measured wall temperatures at the two peep-hole elevations, the outlet temperature and the flue-gas outlet temperature (error bars $\pm 1.96\sigma$), and the parity of predicted against measured temperatures (dotted lines ± 10 K). Source: latham2008_plant_cases.csv, latham_fit.yaml, v2.

354 The bed-side coefficient is validated instead on the plant data, where the
 355 furnace model supplies the outer-wall condition and f_{htg} is a free parameter.
 356 Fig. 3 and Tables 2–3 summarise the calibration. The adopted fit (α_{top} fixed)
 357 gives $F_{gt} = 0.323$, $L_q = 0.456$ and $f_{htg} = 1.83$ with tight intervals, a χ^2 of
 358 34.9 on 24 degrees of freedom (reduced χ^2 1.46), and reproduces the outlet
 359 temperatures of all three cases within 4 K, the lower wall temperatures within
 360 1.1 K and the upper wall temperatures within 12.2 K. The four-parameter fit
 361 lowers χ^2 to 30.5 but is degenerate in L_q and α_{top} (Table 3); fixing α_{top} raises
 362 χ^2 by 14 %, more than the 10 % we set as the threshold for preferring the
 363 simpler model, so both are reported and the three-parameter fit is adopted for
 364 all subsequent work because its parameters are identifiable [TODO: confirm
 365 the choice of the alpha-top-fixed configuration for the RQ work]. The heat-
 366 transfer factor of 1.83 corresponds to a bed-side coefficient of about 1408 W

Table 2: Latham plant cases (thesis Appendix H, absolute temperatures) and calibration results: measured inlet and outlet conditions, model errors (predicted minus measured) of the all-cases fit with α_{top} fixed, and leave-one-case-out held-out errors. Plant C1 is excluded from the fit (inconsistent feed flow). Source: `latham2008_plant_cases.csv`, `latham_fit.yaml` (calibration_alpha_top_fixed).

Case	Rate (%)	T_{in} (K)	p_{in} (bar)	Feed (kmol h ⁻¹)	S/C	T_{out} (K)	$T_{w,u}$ (K)	$T_{w,l}$ (K)	ΔT_{out}	$\Delta T_{w,u}$	$\Delta T_{w,l}$	CV $\Delta T_{w,u}$	CV $\Delta T_{w,l}$
Plant A	99	884.5	30.06	23.5	2.89	1105.5	1095.9	1132.0	+3.6	-8.4	+0.2	-12.8	+0.7
Plant B	92	887.0	29.44	21.1	2.79	1107.2	1082.3	1130.8	+2.2	-3.6	-1.1	-5.4	-1.7
Plant C1	96	890.9	30.27	28.7	2.85	1122.0	1111.2	1151.5	-	-	-	-	-
Plant C2	71	879.9	28.09	15.9	2.86	1108.9	1075.7	1126.2	+3.9	+12.2	-0.5	+18.2	-0.1

Per-tube feed; S/C on total hydrocarbon carbon; upper/lower wall temperatures at 3.66 and 8.53 m of 12.5 m; thesis uncertainties: 2 K outlet, 3 K wall, 8 K flue gas.

Table 3: Calibrated furnace and heat-transfer parameters on Plants A, B and C2 with 95 % t -intervals from the Jacobian covariance. The four-parameter fit has α_{top} at its bound with $r(L_q, \alpha_{top}) = 0.985$; the adopted fit fixes $\alpha_{top} = 0.182$. Source: `latham_fit.yaml`.

Fit	Parameter	Value	95 % interval	s.e.
four parameters	F_{gt}	0.316	[0.286, 0.346]	0.014
four parameters	L_q	0.497	[0.329, 0.665]	0.081
four parameters	α_{top}	0.4	[-0.261, 1.061]	0.32
four parameters	f_{htg}	1.74	[1.427, 2.051]	0.15
four parameters	χ^2 / dof	30.5	23	reduced 1.32
α_{top} fixed (adopted)	F_{gt}	0.323	[0.309, 0.338]	0.0072
α_{top} fixed (adopted)	L_q	0.456	[0.434, 0.478]	0.011
α_{top} fixed (adopted)	f_{htg}	1.83	[1.617, 2.052]	0.11
α_{top} fixed (adopted)	χ^2 / dof	34.9	24	reduced 1.46

367 $\text{m}^{-2} \text{K}^{-1}$ at mid-height of the Plant A tube (four-parameter fit), i.e. between
 368 the two literature correlations and four times the value that reproduces the
 369 Xu–Froment figure; the range $300\text{--}3000 \text{ W m}^{-2} \text{K}^{-1}$ is the epistemic span
 370 carried into the uncertainty analysis through f_{htg} .

371 Leave-one-case-out cross-validation (last two columns of Table 2) shows
 372 that the held-out upper-wall temperature is predicted within 5–18 K and
 373 the lower within 2 K. The upper-wall error is systematic: Plant C2, at 71 %
 374 rate, is predicted too hot at the upper elevation, and Plant A too cold.
 375 The upper peep hole lies in the flame zone, where a single parabolic release
 376 profile with a load-independent flame length cannot follow the change in
 377 flame shape with firing rate. This is the main structural limitation of the
 378 furnace model, and it is why the automated tests for the upper-wall error are
 379 recorded as expected failures against the 4.5 K target derived from the stated
 380 measurement uncertainty. The consequence for the conclusions is examined
 381 in Section 3.5.

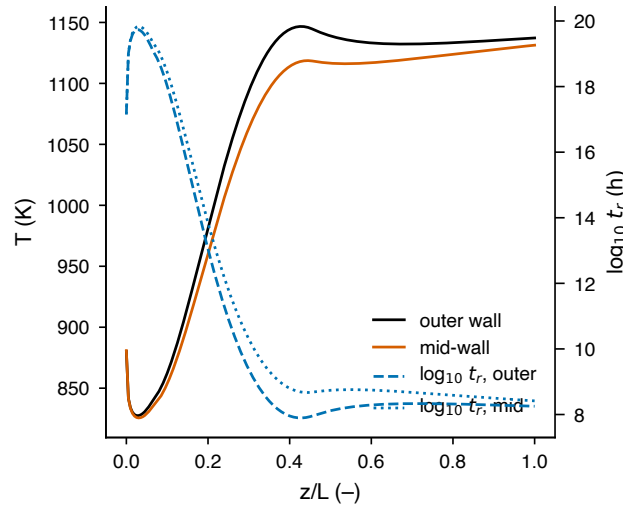


Figure 4: Outer-wall and mid-wall temperature profiles of the calibrated Plant A tube and the corresponding local rupture time from the placeholder Larson–Miller curve at the local hoop stress; the hot spot [TODO: state the hot-spot elevation] governs the life. Source: latham_fit.yaml (creep_life_first_estimate), yeh2021_manaurite_xm.yaml, v2.

382 Fig. 4 shows the life profile of the calibrated Plant A tube. The hot spot
 383 lies at about [TODO: quote z/L of the hot spot from the figure or the YAML]
 384 of the heated length, where the outer wall reaches 1147 K at a hoop stress of
 385 12.9 MPa; the rupture time from the placeholder minimum curve is 10^8 h at

386 this point. This absolute value is not credible as a tube life and illustrates
 387 why the placeholder curve must be replaced: the manufacturer's curve carries
 388 no data below about 20 MPa and the low hoop stress of a thick tube at 30 bar
 389 places the operating point in the extrapolated region [TODO: comment on
 390 the stress range of the placeholder curve and the extrapolation]. All results
 391 below are therefore expressed as ratios of consumption rates, which depend
 392 on the slope of the master curve at the operating point rather than on its
 393 absolute position.

394 3.2. RQ1: what governs hot-spot temperature and life

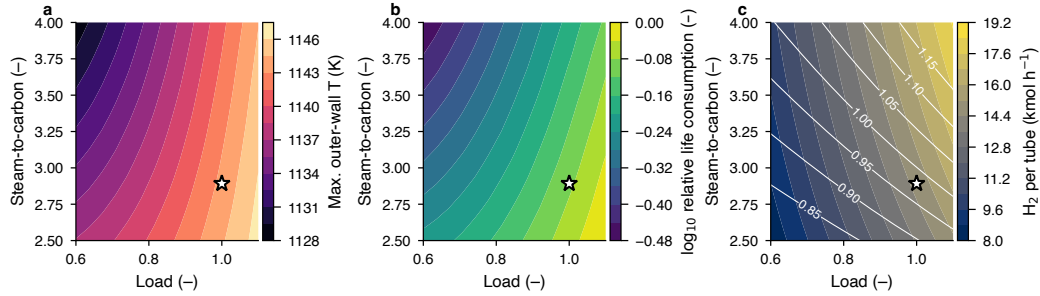


Figure 5: Closed-loop operating maps in load and steam-to-carbon ratio at constant outlet temperature (1105.55 K): (a) hot-spot temperature, (b) $\log_{10}$ of the life consumption relative to the base case and (c) net hydrogen per tube with contours of the firing factor required. The star marks the calibrated Plant A base. Source: data/lhs_runs/grid_sc_load_v2.csv.gz, v2.

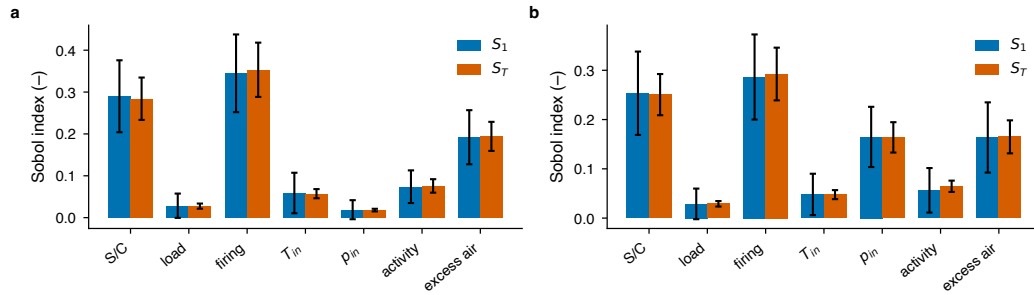


Figure 6: First-order (S_1) and total-effect (S_T) Sobol indices with 95 % confidence intervals for (a) the hot-spot temperature and (b) $\log_{10} t_r$ at the hot spot, from 4096 physics evaluations of a Saltelli design over the seven operating inputs. Source: data/design/sobol_v2.json, v2.

Table 4: Sobol first-order (S_1) and total-effect (S_T) indices with 95 % confidence half-widths from a Saltelli design ($N = 256$, 4096 physics runs) over the v2 operating space. Source: `data/design/sobol_v2.json` (v2).

Input	$S_1 (T_{wo,max})$	$S_T (T_{wo,max})$	$S_1 (\log_{10} t_r)$	$S_T (\log_{10} t_r)$	$S_1 (\text{CH}_4 \text{ slip})$	$S_T (\text{CH}_4 \text{ slip})$
steam-to-carbon	0.29 ± 0.09	0.28 ± 0.05	0.25 ± 0.08	0.25 ± 0.04	0.04 ± 0.03	0.04 ± 0.01
load	0.03 ± 0.03	0.03 ± 0.01	0.03 ± 0.03	0.03 ± 0.01	0.19 ± 0.07	0.19 ± 0.03
firing	0.34 ± 0.09	0.35 ± 0.06	0.29 ± 0.09	0.29 ± 0.05	0.54 ± 0.12	0.54 ± 0.10
inlet T	0.06 ± 0.05	0.06 ± 0.01	0.05 ± 0.04	0.05 ± 0.01	0.08 ± 0.06	0.09 ± 0.02
inlet p	0.02 ± 0.02	0.02 ± 0.00	0.16 ± 0.06	0.16 ± 0.03	0.03 ± 0.03	0.03 ± 0.01
activity	0.07 ± 0.04	0.08 ± 0.02	0.06 ± 0.05	0.06 ± 0.01	0.00 ± 0.01	0.00 ± 0.00
excess air	0.19 ± 0.06	0.19 ± 0.03	0.16 ± 0.07	0.16 ± 0.03	0.11 ± 0.04	0.11 ± 0.02

The closed-loop maps (Fig. 5) show the structure of the operating space when the outlet temperature is held by the firing controller. At constant outlet temperature, the hot-spot temperature rises with load because the duty per tube rises while the furnace-gas temperature at the flame zone is fixed by the fuel-to-feed ratio, and it falls with steam-to-carbon ratio because the endothermic load moves towards the tube inlet [TODO: check this physical reading of panel (a)]. The life map (panel b) is the image of the temperature map through the Larson–Miller relation and spans about two decades across the space. Hydrogen production (panel c) increases with both load and steam-to-carbon ratio, the latter through the shift of equilibrium, at the cost of a higher firing factor.

The Sobol analysis over all seven inputs (Fig. 6, Table 4) ranks the drivers of the hot-spot temperature: the specific firing factor ($S_T = 0.35$), the steam-to-carbon ratio (0.28) and the excess air (0.19) together explain 82 % of the variance; inlet temperature, catalyst activity and load contribute below 0.1 each, and inlet pressure is negligible. Excess air matters because at fixed fuel it lowers the adiabatic flame temperature and raises the flue-gas flow, both of which move heat release down the tube. The ranking for $\log_{10} t_r$ is the same except that the inlet pressure enters with $S_T = 0.16$ through the hoop stress, which is invisible in the temperature. First-order and total-effect indices agree within their confidence intervals for every input, so interactions are weak over the operating space and the influence of each variable can be read from one-at-a-time maps. The methane slip is governed by the firing factor ($S_T = 0.54$) and the load, as expected for an approach-to-equilibrium quantity.

The surrogates that make the following campaigns possible are summarised in Table 5 and Fig. 7. The Gaussian process is selected for seven

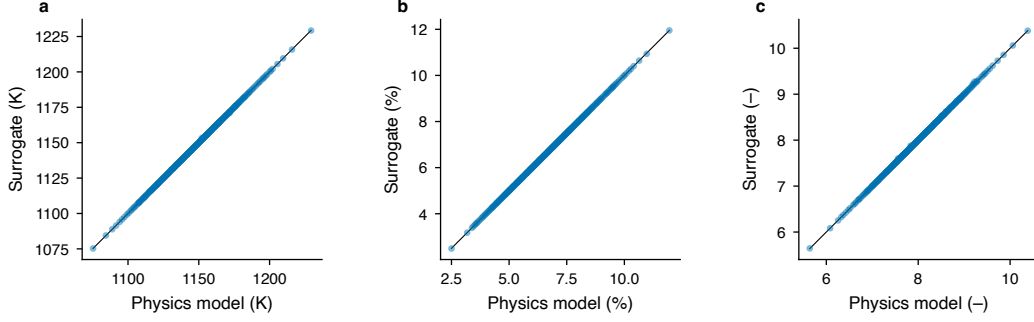


Figure 7: Parity of the selected surrogates against the physics model on the independent Saltelli test set for (a) the hot-spot temperature, (b) dry methane slip and (c) $\log_{10} t_r$ at the hot spot, with nominal 90 % CV+ conformal intervals. Source: `data/design/surrogate_metrics_v2.json`, v2.

Table 5: Best surrogate per target ($N_{train} = 2000$ LHS, $N_{test} = 4096$ Saltelli points), test R^2 and RMSE, 5-fold CV RMSE, and empirical coverage (PICP) and mean width (MPIW) of the nominal 90 % CV+ conformal interval. Source: `data/design/surrogate_metrics_v2.json`.

Target	Unit	Model	R^2	RMSE	CV RMSE	PICP ₉₀	MPIW ₉₀
T_wo_max_K	K	gp__physics	1	0.113	0.133	0.957	0.25
z_frac_T_wo_max	–	mlp__base	0.6436	0.0319	0.0522	0.971	0.0499
T_out_K	K	gp__physics	1	0.0182	0.0254	0.940	0.0614
P_out_bar	bar	gp__base	1	0.000287	0.000563	0.961	0.000998
CH4_slip_dry_pct	%	gp__physics	1	0.00125	0.00142	0.953	0.00398
H2_net_kmol_h	kmol h ⁻¹	gp__base	1	0.000847	0.00116	0.946	0.00286
duty_W	W	gp__base	1	3.9e+03	4.14e+03	0.949	1.21e+04
log10_t_r_hot	–	gp__base	0.9999	0.00566	0.00597	0.959	0.0111

of the eight targets, and the physics features improve the hot-spot temperature, outlet temperature and slip. Test RMSE is 0.11 K for the hot-spot temperature and 0.0057 decades for $\log_{10} t_r$, two orders of magnitude below the calibration residuals, at a speed-up of 4658 relative to the physics model. The CV+ intervals over-cover: at a nominal level of 0.90 the empirical coverage is 0.94–0.97 for all targets (0.957 for the hot-spot temperature). Over-coverage is expected for CV+, whose guarantee is one-sided, and is conservative; the interval widths are nevertheless small enough (0.25 K for the hot-spot temperature) that surrogate error is immaterial to the conclusions.

3.3. RQ2: life cost of flexible operation and ageing

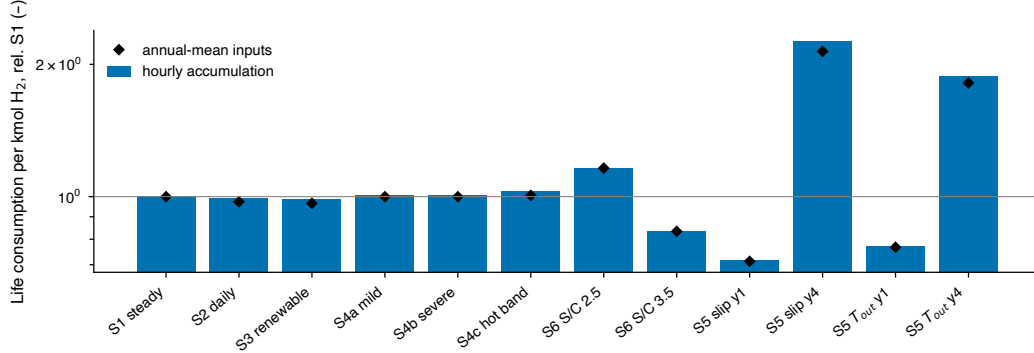


Figure 8: Annual life consumption per kmol of hydrogen relative to the steady base S1 for the RQ2 scenarios, from hourly accumulation (bars) and from a single evaluation at the annual-mean inputs (diamonds). Source: `data/scenarios/summary_v2.csv`, v2.

Table 6 and Fig. 8 answer RQ2 for one year of operation at the calibrated base. Under the outlet-temperature controller, load-following costs almost nothing in tube life per unit of hydrogen: the daily profile (S2) and the renewable-following profile (S3) consume 0.991 and 0.983 of the S1 life per kmol H_2 , i.e. slightly *less*, because the hot-spot temperature at constant outlet temperature falls with load (Fig. 5a) and the hours at reduced load more than compensate the hours at overload. Hydrogen output falls by 10 % and 15 % respectively, so the absolute annual damage falls in proportion. A steady-state assessment at the annual-mean inputs reproduces the hourly accumulation to within 2 % for every scenario without discrete events (ratio 0.982 for S3), which means that the non-linearity of the Larson–Miller relation is not strong enough over the load range of this controller to make

Table 6: RQ2 scenario summary for one year (8760 h) of the Plant A twin at 21.6 % excess air: hydrogen per tube, Robinson damage D , life consumption per kmol H_2 relative to S1, ratio of the annual-mean-condition damage estimate to the hourly-accumulated damage, mean and maximum hot-spot temperature and mean firing factor. Life uses the Yeh Manaurite XM placeholder curve. Source: `data/scenarios/summary_v2.csv`.

Scenario	H_2 (10^3 kmol)	D (-)	Rel. life cost	Avg.-cond. ratio	$\bar{T}_{wo,max}$ (K)	max $T_{wo,max}$ (K)	Firing
S1 steady base	128.5	9.08e-05	1.000	1.000	1143	1143	0.978
S2 daily load-following	115.7	8.1e-05	0.991	0.982	1141	1143	0.944
S3 renewable-following	109.7	7.61e-05	0.983	0.982	1140	1146	0.927
S4a mild over-firing (48 h)	128.6	9.14e-05	1.007	0.993	1144	1157	0.979
S4b severe over-firing (6 h)	128.5	9.11e-05	1.004	0.996	1143	1179	0.978
S4c hot band +40 K (24 h)	128.5	9.33e-05	1.028	0.980	1144	1183	0.978
S6 steam-to-carbon 2.5	119.9	9.83e-05	1.161	1.000	1144	1144	0.897
S6 steam-to-carbon 3.5	139.8	8.23e-05	0.834	1.000	1142	1142	1.092
S5 ageing, hold slip, year 1	128.5	6.49e-05	0.715	0.997	1138	1139	0.974
S5 ageing, hold slip, year 4	128.5	0.000205	2.255	0.949	1156	1164	0.989
S5 ageing, hold T_{out} , year 1	129.0	7e-05	0.768	0.998	1139	1140	0.982
S5 ageing, hold T_{out} , year 4	127.4	0.000169	1.875	0.967	1153	1159	0.969

time-resolved simulation necessary for smooth load variations [TODO: discuss whether this conclusion holds for a plant without outlet-temperature control].

Discrete over-firing events are different in kind. During a mild event (10 % firing) the damage rate is 2.4 times the base rate, during a severe event (25 %) it is 6.9 times, and a 40 K hot band raises it by a factor of 11.1; but because the events are short, their share of the annual damage is 0.4–2.7 % (2.7 % for the hot band) and the annual life cost per kmol rises by at most 3 %. The annual-mean evaluation underestimates the event scenarios (diamonds below bars in Fig. 8), as it must for a convex damage law, but the error is small at these event frequencies. Steam-to-carbon ratio has a larger steady effect: operating at 2.5 costs 1.16 and at 3.5 costs 0.83 of the S1 life per kmol H_2 , because the higher ratio both cools the hot spot and raises the hydrogen yield.

Catalyst ageing dominates every operational effect. With the activity falling from 0.30 to 0.10 over four years and the controller holding the methane slip, the life cost per kmol H_2 rises from 0.71 in the first year (fresh catalyst) to 2.25 in the fourth as the hot spot climbs from 1137 to 1164 K; holding the outlet temperature instead gives 1.88 in the fourth year because the slip is allowed to rise. Over the campaign the life consumed in year 4 is three times that in year 1, and the question of when to change the catalyst is therefore also a tube-life question [TODO: comment on whether the activity range 0.30–0.10 is representative of a campaign].

467 3.4. RQ3: Pareto-optimal regimes and the steam credit

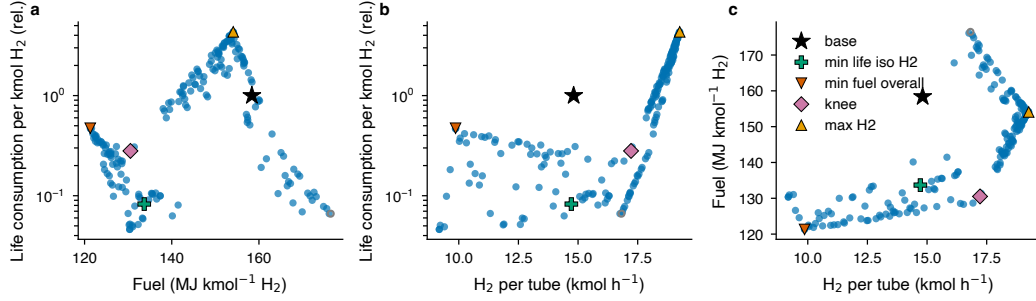


Figure 9: Physics-verified NSGA-II Pareto set in (a) firebox fuel per kmol H_2 against life consumption per kmol H_2 , (b) net hydrogen per tube against life consumption and (c) hydrogen against fuel; open circles are members infeasible under the physics model, and the marked regimes are listed in Table 7. Source: `data/optimization/pareto_v2.csv`, `regimes_v2.csv`, v2.

Table 7: RQ3 representative regimes (physics-verified) from the NSGA-II Pareto set (200×300 generations; 199 of 200 members feasible under the physics model). Fuel = firebox fuel per kmol H_2 ; life = consumption rate per kmol H_2 relative to the base (Yeh placeholder curve). Source: `data/optimization/regimes_v2.csv`.

Regime	Load	S/C	T_{in} (K)	Excess air (%)	Firing	H_2 (kmol h^{-1})	Fuel (MJ kmol $^{-1}$ H_2)	Rel. life cost	$T_{wo,max}$ (K)	CH_4 slip (%)
base	1.00	2.89	885	21.6	1.000	14.82	158.4	1.000	1147	7.81
min life iso H_2	0.96	4.00	895	17.5	0.873	14.72	133.7	0.083	1109	7.77
knee	1.10	4.00	900	5.0	0.871	17.22	130.5	0.280	1130	7.15
max H_2	1.10	4.00	900	5.0	1.150	19.26	154.1	4.329	1177	3.89
min fuel overall	0.60	4.00	900	5.0	0.850	9.87	121.3	0.472	1128	5.72
min life overall	0.60	4.00	858	24.9	0.850	9.18	130.4	0.046	1091	7.81

468 The Pareto set (Fig. 9, Table 7) contains 200 members, of which 199
469 remain feasible under the physics model; the maximum discrepancy be-
470 tween surrogate and physics on the set is 0.03 in relative life and below
471 0.1 MJ kmol $^{-1}$ in fuel. The calibrated base is dominated by 37 members.
472 The knee of the front raises hydrogen by +16 % while cutting fuel per kmol
473 by -18 % and life consumption per kmol to 0.28 of the base; it does so by
474 running at 110 % load with the minimum excess air, the maximum inlet tem-
475 perature and a lower firing factor. At constant production (iso- H_2 regime),
476 the life cost can be reduced to 0.083 of the base with -16 % fuel by reducing
477 the firing and letting the hot spot fall from 1147 to 1109 K. The maximum-
478 hydrogen corner (+30 %) costs 4.3 times the base life per kmol and runs close
479 to the wall-temperature limit.

480 Every member of the front sits at the upper bound of the steam-to-carbon
 481 ratio (3.96–4.00). This is an artefact of the objective: with firebox fuel alone,
 482 additional steam is free, and the equilibrium shift and the cooling of the hot
 483 spot both favour maximum steam. The steam-credit sweep (Fig. 10, Table 8)
 484 shows how the answer changes when steam raising and feed preheat are
 485 charged. With steam fully charged ($\alpha = 0$) the minimum-fuel regime moves
 486 to a steam-to-carbon ratio of 2.66 and the knee to 3.3, while the minimum-
 487 life regime stays at 4.0 but now costs +3.2% more total fuel than the base
 488 for a life cost of 0.059. For $\alpha \geq 0.5$, i.e. when at least half of the steam duty
 489 is recovered from the flue gas or credited as export steam, the ratio of 4.0
 490 is optimal for all three regimes. The plant’s own steam balance therefore
 491 decides whether tube life and fuel economy pull in the same direction or
 492 against each other [TODO: relate the steam-credit range to a typical SMR
 493 steam balance with waste-heat boiler and export steam].

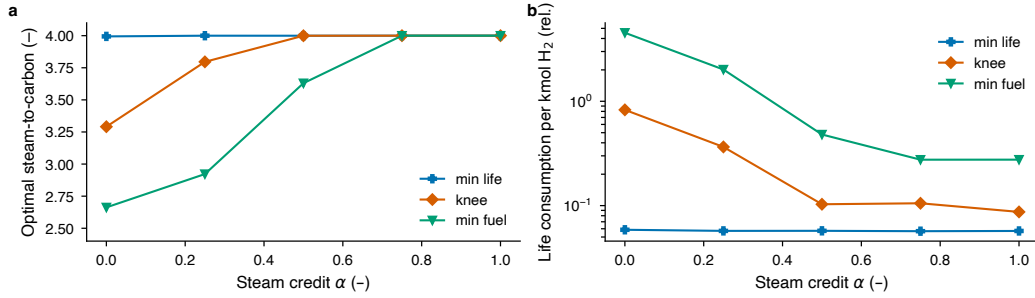


Figure 10: Iso-production front as a function of the steam credit α : (a) optimal steam-to-carbon ratio and (b) life consumption per kmol H_2 relative to the base for the minimum-life, knee and minimum-fuel regimes. Source: data/optimization/steam_credit_v2.json, v2.

494 3.5. RQ4: uncertainty and robustness

495 Table 9 and Fig. 11 give the propagated uncertainty. At the base point
 496 the 90 % interval of the hot-spot temperature is 42 K wide and that of $\log_{10} t_r$
 497 spans 7.04–8.73, i.e. a factor of 50 in absolute rupture time. The paired ratios
 498 are much tighter: the knee regime costs 0.20–0.64 of the base life per kmol
 499 and the fourth ageing year 1.8–9.0 times, so the ranking of regimes survives
 500 an uncertainty that makes the absolute life meaningless. The variance decomposition
 501 lies inside the 90 % interval at every point. The variance decomposition
 502 (Table 10, Fig. 12) attributes 62% of the variance of $\log_{10} t_r$ to the creep

Table 8: Steam-credit sweep of the iso-production (H_2 = base ± 1 %) life-fuel front: $\alpha = 1$ counts firebox fuel only, $\alpha = 0$ charges steam raising and feed preheat in full. Physics-verified. Source: `data/optimization/steam_credit_v2.json`.

α	Regime	S/C	Load	Firing	Excess air (%)	Rel. life cost	Total fuel vs base (%)	$T_{wo,max}$ (K)
0.00	min life	3.99	0.96	0.909	24.6	0.059	+3.2	1103
0.00	knee	3.29	0.97	0.850	5.1	0.828	-6.6	1144
0.00	min fuel	2.66	1.00	0.888	5.0	4.539	-8.6	1171
0.25	min life	4.00	0.96	0.943	24.9	0.057	+2.3	1103
0.25	knee	3.80	0.94	0.850	5.0	0.365	-7.1	1131
0.25	min fuel	2.92	0.99	0.865	5.0	2.016	-8.9	1158
0.50	min life	4.00	0.96	1.039	25.0	0.057	+5.5	1103
0.50	knee	4.00	0.95	0.850	13.8	0.103	-7.6	1112
0.50	min fuel	3.63	0.95	0.850	5.0	0.480	-10.5	1135
0.75	min life	4.00	0.96	0.911	25.0	0.057	-6.6	1103
0.75	knee	4.00	0.96	0.850	13.5	0.105	-12.2	1112
0.75	min fuel	4.00	0.93	0.850	5.0	0.276	-14.6	1127
1.00	min life	4.00	0.96	0.958	25.0	0.057	-7.3	1103
1.00	knee	4.00	0.96	0.850	15.3	0.087	-17.7	1109
1.00	min fuel	4.00	0.93	0.850	5.0	0.276	-20.4	1127

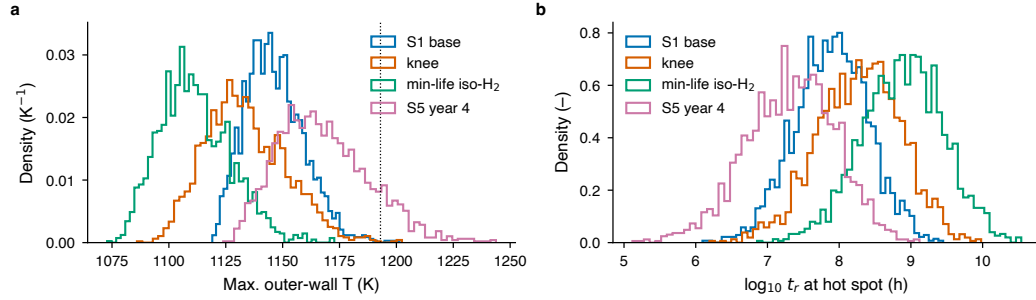


Figure 11: Monte Carlo distributions (2048 samples of all uncertainty groups through the physics model) of (a) the hot-spot temperature, with the 1193K limit, and (b) $\log_{10} t_r$ at the hot spot, at the S1 base, the RQ3 knee, the iso-H₂ minimum-life regime and the end of the fourth ageing year. Source: `data/uq/mc_v2_*.csv.gz`, v2.

Table 9: RQ4 Monte Carlo (scrambled Sobol design, $N = 2048$, all uncertainty groups) through the physics model: median and 5–95 % interval of the hot-spot temperature, of $\log_{10} t_r$ and of the paired life consumption per kmol H_2 relative to the base. Source: `data/uq/uq_summary_v2.json`.

Operating point	$T_{wo,max}$ (K)	$\log_{10} t_r$ (h)	Rel. life cost
S1 base	1145 [1127, 1169]	7.91 [7.04, 8.73]	1.000 [1.000, 1.000]
RQ3 knee	1132 [1107, 1165]	8.30 [7.33, 9.20]	0.338 [0.201, 0.643]
RQ3 min-life iso- H_2	1110 [1088, 1138]	8.90 [7.94, 9.80]	0.100 [0.069, 0.159]
S5 year 4, hold slip	1165 [1138, 1203]	7.35 [6.28, 8.30]	3.478 [1.841, 8.974]
S/C 2.5	1146 [1129, 1166]	7.88 [7.04, 8.69]	1.153 [0.916, 1.350]
S/C 3.5	1144 [1127, 1172]	7.94 [7.04, 8.83]	0.846 [0.587, 1.119]
load 0.70	1138 [1123, 1156]	8.08 [7.22, 8.90]	0.966 [0.638, 1.350]
load 0.85	1141 [1124, 1162]	8.00 [7.14, 8.82]	0.951 [0.787, 1.143]

Table 10: Variance decomposition at the S1 base point: one-group-at-a-time share of the joint variance ($N = 1024$ per group) and group total-effect Sobol index from a Gaussian-process surrogate of the joint sample. Source: `data/uq/uq_summary_v2.json`.

Group	share $\log_{10} t_r$	$S_T \log_{10} t_r$	share $T_{wo,max}$	$S_T T_{wo,max}$
kinetics	0.233	0.225	0.513	0.620
heat transfer	0.050	0.114	0.111	0.302
creep	0.622	0.655	0.211	0.093
measurement	0.094	0.036	0.202	0.092
sum / interaction residual	0.999 / +0.001		1.037 / -0.037	

Table 11: Robustness of the RQ2/RQ3 conclusions: fraction of paired Monte Carlo samples supporting each statement, with the calibrated flame length L_q and with the structural variant $L_q \text{load}^{0.5}$. Source: `data/uq/uq_summary_v2.json`.

Statement	nominal L_q	$L_q \propto \text{load}^{0.5}$
ageing year 4 costs $> 2 \times$ S1 life per kmol	0.906	0.906
knee regime $< 0.5 \times$ base	0.806	0.936
S/C 3.5 cheaper in life than S/C 2.5	0.836	0.836
S2 daily $\leq$ S1	0.568	0.000
S3 renewable $\leq$ S1	0.632	0.001

group, dominated by the heat-to-heat scatter and the Larson–Miller constant, 23 % to the kinetics, 9 % to the wall-temperature measurement and only 5 % to the calibrated heat-transfer parameters; for the hot-spot temperature the kinetics group dominates (51 %), followed by the measurement offset and the wall thickness. The one-at-a-time shares sum to within 0.04 of one, and the Sobol total effects on the surrogate confirm the ranking, so the groups act almost additively.

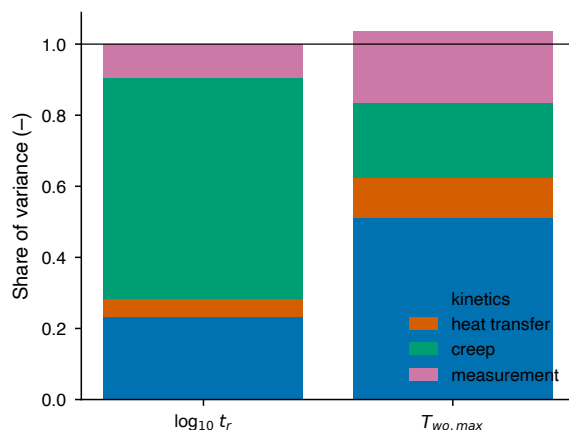


Figure 12: One-group-at-a-time shares of the variance of $\log_{10} t_r$ and of the hot-spot temperature at the S1 base point. Source: data/uq/uq_summary_v2.json, v2.

The robustness fractions (Table 11) state which conclusions hold across the joint uncertainty. That the fourth ageing year costs more than twice the S1 life per kmol holds in 0.91 of the samples, that the knee regime halves the life cost in 0.81, and that steam-to-carbon 3.5 is cheaper in life than 2.5 in 0.84; these three conclusions are robust and, for the knee, become more so under the flame-length variant. The conclusion that flexible operation costs no life per kmol is not robust: it holds in 0.63 of the samples with the calibrated flame length, close to a coin toss because the nominal effect is within 2 % of unity, and it is reversed in all samples when the flame shortens with load, under which the hot spot at 70 % load moves up the tube into the flame zone and the life cost per kmol rises to 1.71 times the base. The sign of the flexible-operation effect is therefore determined by a structural feature of the furnace, the response of the flame length to load, that the three plant cases cannot identify.

524 4. Discussion

525 4.1. Industrial implications

526 Three results have direct operational consequences. First, the drivers of
527 tube life are the firing intensity, the steam-to-carbon ratio and the excess air,
528 in that order, and they act almost independently (Section 3.2). Of these,
529 excess air is the least appreciated: at fixed fuel it lowers the hot-spot tem-
530 perature by shifting heat release down the tube, and the optimiser exploits
531 it in both directions, choosing minimum excess air when hydrogen is the pri-
532 ority and maximum excess air when life is. Excess air is also the cheapest
533 variable to move on an operating plant, and its effect can be verified with
534 the existing pyrometer survey [TODO: comment on the flue-gas oxygen and
535 efficiency penalty of higher excess air].

536 Second, the trade-off between production, fuel and life has a knee that
537 is far from the current operating point of the calibrated plant. The base is
538 dominated by a substantial part of the front, and the knee regime obtains
539 more hydrogen at less fuel per kmol and a fraction of the life cost, mainly by
540 running at higher load with lower specific firing and the highest inlet temper-
541 ature. Whether these gains are realisable depends on constraints outside the
542 twin (preheat-coil metallurgy at 900 K inlet temperature, PSA capacity at
543 110 % load, downstream steam balance), and the steam-credit sweep shows
544 that the recommended steam-to-carbon ratio moves from 4.0 to about 2.7
545 depending on how the plant values steam. The twin cannot decide this, but
546 it makes the dependence explicit, which is the useful output: the plant's
547 steam balance, not the reformer, determines whether running steam-rich is
548 a life-saving measure or a fuel penalty.

549 Third, the life cost of load-following is small compared with the life cost of
550 catalyst ageing, provided the outlet temperature is controlled. The practical
551 message is that a plant that must follow demand should not fear the tube-life
552 consequences of smooth load changes, but should watch the end-of-campaign
553 hot-spot temperature under a methane-slip target, where the twin predicts
554 the fourth year to cost 2.3 times the first per kmol of hydrogen. Under
555 uncertainty this ageing conclusion is the most robust of the study, and the
556 conclusion on load-following the least. The difference between the two comes
557 from the size of the nominal effect (a factor of 2.3 against 2 %), not from a
558 difference in modelling quality, and it is an argument for reporting robustness
559 fractions alongside nominal results in any digital-twin study [TODO: soften
560 or strengthen this recommendation].

561 4.2. The flame-length response to load as a measurement need

562 The one structural assumption that changes a conclusion is how the heat-
563 release profile of the burners responds to the firing rate. The calibrated model
564 uses a single dimensionless flame length, and the cross-validation residuals
565 at the upper peep hole (Section 3.1) already show that this is the model’s
566 weakest element: the partially loaded Plant C2 is predicted too hot at the top
567 and the fully loaded Plant A too cold. When the flame is allowed to shorten
568 with load, the flexible-operation conclusion reverses because the hot spot at
569 reduced load moves into a hotter part of the furnace. The three published
570 operating points cannot separate the two hypotheses. What would settle the
571 question is a pyrometer survey at three or four elevations at two or three
572 firing rates on one furnace, which most plants already perform in part during
573 turnaround planning; a digital twin of the kind presented here turns such
574 a survey into an estimate of the flame-length response with its uncertainty,
575 and the robustness fractions would then no longer straddle 0.5. We regard
576 this as the single most valuable measurement for tube-life-aware operation
577 of top-fired reformers [TODO: check against burner-manufacturer data on
578 flame length versus firing rate for the burner type].

579 4.3. Limitations

580 The model represents one average tube. Tube-to-tube variation in a top-
581 fired furnace, caused by burner imbalance, row position and flow maldistri-
582 bution, is known to be of the order of 20 K in wall temperature [TODO: cite a
583 survey of tube-to-tube wall temperature spread], which by the Larson–Miller
584 relation corresponds to a factor of two to three in rupture time between the
585 hottest and the average tube. The measurement group of the uncertainty
586 analysis (± 10 K) partly covers this, and a per-tube extension would replace
587 the single wall-temperature offset by a distribution over the tube population,
588 but the present results are statements about an average tube.

589 The life model covers creep only. Thermal fatigue during start-up and
590 shutdown, creep–fatigue interaction, carburisation and the relaxation of ther-
591 mal stresses across the wall are not modelled, and the hoop stress is the pres-
592 sure stress alone; the thermal stress from the radial temperature gradient,
593 which is of comparable magnitude in thick tubes [TODO: quantify the ther-
594 mal stress for this wall thickness and flux], is neglected. The scenario results
595 should therefore be read as the creep-life cost of the scenarios, not their total
596 life cost, and the ranking of flexible-operation scenarios could change if the
597 number of thermal cycles were counted.

598 The rupture curve is a placeholder. The manufacturer’s minimum curve
 599 for a proprietary alloy reproduced by Yeh [18] was used because no public
 600 curve with scatter information was available in the transcribed form the
 601 module requires, and the absolute rupture times it yields at the low hoop
 602 stress of this tube are not credible. The uncertainty analysis compensates
 603 with a factor-of-two heat-to-heat scatter and a ± 1 range on the Larson–
 604 Miller constant, and the creep group is the largest contributor to the variance
 605 of $\log_{10} t_r$; replacing the curve by a heat-normalised fit to the NIMS data
 606 sheets for the 25Cr–35Ni–Nb alloys [11, 12] is the first step for any absolute
 607 statement. The module is built for this replacement and the ingestion has
 608 been tested on synthetic data.

609 The scenarios are quasi-steady: each hour is a steady state of the twin,
 610 and the thermal inertia of the tube, the catalyst and the furnace refractory
 611 is neglected. For load ramps of 0.1 per hour and events of one hour or
 612 longer this is a reasonable approximation [TODO: compare the tube and
 613 refractory thermal time constants with the ramp rates], but it excludes the
 614 transient overshoot of the wall temperature during a fast firing step, which
 615 would increase the cost of the over-firing events. Finally, the calibration rests
 616 on three operating points of one furnace from a single published source, so
 617 the parameter intervals are conditional on that furnace and the transfer to
 618 another reformer requires recalibration on its own survey data; the pipeline
 619 is designed so that this is a matter of replacing one CSV file.

620 5. Conclusions

621 A physics-informed digital twin of an industrial top-fired steam methane
 622 reformer has been built from published kinetics, plant data and creep curves,
 623 coupling a one-dimensional tube model, a long-furnace model, a Larson–
 624 Miller life module, Gaussian-process surrogates with conformal prediction
 625 intervals and Monte Carlo uncertainty propagation in one open, versioned
 626 pipeline.

- 627 • The tube model reproduces the industrial case of Xu and Froment
 628 within 14.0 K in gas temperature once the film coefficient is matched,
 629 and the heat-transfer correlations printed in that paper are found to
 630 overpredict the film coefficient by an order of magnitude at industrial
 631 Reynolds numbers. The coupled model calibrated on three plant oper-
 632 ating points reproduces outlet temperatures within 4 K and wall tem-
 633 peratures within 12 K, with identifiable parameters and a systematic

- 634 residual at the upper elevation that points to the flame-length response
635 to load.
- 636 • Firing intensity, steam-to-carbon ratio and excess air explain 82 % of
637 the variance of the hot-spot temperature with negligible interactions;
638 inlet pressure enters the life only through the hoop stress.
 - 639 • Under outlet-temperature control, load-following and renewable-following
640 cost less than 2 % of the steady life per kmol of hydrogen, short over-
641 firing events at most 3 %, and steam-to-carbon ratio ± 17 %; catalyst
642 ageing under a methane-slip target multiplies the life cost by 2.25 in
643 four years.
 - 644 • The calibrated base is Pareto-dominated: a knee regime gives +16 %
645 hydrogen at -18 % fuel per kmol and 0.28 of the life cost, and at constant
646 production the life cost can be cut to 0.083. The optimal steam-to-
647 carbon ratio depends entirely on the steam credit, moving from 4.0 to
648 2.7 when steam raising is charged in full.
 - 649 • The 90 % uncertainty on the absolute rupture time spans a factor of
650 50, dominated by creep-curve scatter (62 %), while paired ratios be-
651 tween regimes remain tight. The ageing, knee and steam-to-carbon
652 conclusions hold in 81 % to 91 % of the samples; the flexible-operation
653 conclusion does not survive a load-dependent flame length.

654 The next steps are the transcription of the NIMS rupture data for the
655 HP-Nb alloys into the prepared ingestion path, a per-tube extension with
656 a measured wall-temperature distribution, and, above all, a multi-elevation
657 pyrometer survey at several firing rates to identify the flame-length response
658 that decides the life cost of flexible operation.

659 Data and code availability

660 The model code (`rdt` package), the pipeline entry point (`python -m`
661 `rdt.pipeline -stage all -version v2`), the transcribed literature data
662 with their provenance annotations, the digitised verification curves, all v2 re-
663 sult files and the scripts that generate every figure, table and number in this
664 manuscript are available at [\[TODO: GitHub URL\]](#) and archived at [\[TODO: Zenodo DOI\]](#). The literature PDFs are not redistributed. The two Monte
665 Carlo result files above 5 MB are provided in the archive only. Transcribed
666 data are released under CC BY 4.0 and code under the MIT licence.

668 CRediT authorship contribution statement

669 **Izbassar Orynbassar:** Conceptualisation, Methodology, Software, Val-
670 idation, Formal analysis, Data curation, Visualisation, Writing – original
671 draft, Writing – review and editing. **[TODO: Second author]:** [TODO:
672 Supervision, Resources, Validation of the chemical-engineering content, Writ-
673 ing – review and editing].

674 Declaration of generative AI and AI-assisted technologies in the 675 writing process

676 During the preparation of this work the authors used Claude (Anthropic)
677 to assist with code development, data transcription checks, figure and table
678 generation and drafting of the manuscript text. After using this tool, the
679 authors reviewed and edited the content as needed and take full responsibility
680 for the content of the published article [TODO: confirm the wording required
681 by the journal at submission].

682 Declaration of competing interest

683 The authors declare that they have no known competing financial inter-
684 ests or personal relationships that could have appeared to influence the work
685 reported in this paper. M.S. is employed in the industrial gas sector; the
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